Q#05:-

Buffer Capacity = 50,000

Packet size = 1000 bytes

Buffer size = 50 MB

Outgoing link = 100 Mbps

= 100,000,000 b/s

Arrival rate during burst = 150 Mbps

= 150,000,000 b/s

burst duration = 10s

1) Net excess rate = arrival - outgoing

= 50,000,000 b/s ÷ 8

= 6250000 bytes/s

Time to fill buffer = 50,000,000 ÷ 6250000

= 8 seconds

2) Remaining burst time ⇒ 10 - 8 = 2 sec

50000000 x 2 = 100000,000 bits ÷ 8 = 12500,000 bytes

Number of packets dropped

12500,000 ÷ 1000 = 12500packets

3) Active Queue Management

↳ continuously monitors queue length or delay packets before buffer overflow. Maintain low latency and avoid sudden bursts of packet loss.

Congestion Feedback and Traffic Policing

↳ Adjust their rates smoothly while traffic shaping regulates data flow to prevent sudden spikes.

Q#04:-

$R = 350,000,000$ b/s

$L = 250,000$ bits

$\lambda = 35$ req/s

Average Internet = 2.5

$S = L/R = 250,000 / 350,000,000$

$= 7.1429 \times 10^{-4}$ sec

$\mu = 1/S \cong 1400$ requests/sec

$\rho = \lambda * S = 35 \times 0.000714285 7$

$= 0.025$

1)

$T_{access} = S/(1-\rho) = 7.3244 \times 10^{-4}$ sec

$T_{total} = T_{access} + T_{Internet} = 0.0007 3244 + 2.5$

$= 2.5$ seconds

miss rate = 0.7

$T_{cache} = 0.01$ sec

$\lambda_{miss} = 0.7 \times 35 = 24.5$ req/s

$\rho_{miss} = \lambda_{miss} * S = 24.5 \times 0.00071428$

$= 0.0175$

$T_{access_miss} = S/(1-\rho_{miss}) = 0.0007275$ sec

$T_{miss} = T_{access_miss} + T_{Internet} = 0.0007275 + 2.5$

$= 2.5$ sec

$T_{avg} = 0.7(T_{miss}) + (0.3)T_{cache} = 0.7 \times 2.5 + 0.3 \times 0.01$

$T_{avg} = 1.75 + 0.0925 + 0.003$

$= 1.7525$ second.